$$U_{xy} \neq U_{yx}$$

if analytic $\xrightarrow{\text{true}}$ harmonic + CR satisfy

but

harmonic + CR satisfy $\xrightarrow[\text{true}]{\text{not}}$ analytic

$$u(x,y) + i\,v(x,y)$$

example
[Slide pg 4]

check if harmonic?

Do U_x
then U_{xx}

now, do U_y
then U_{yy}

yes?

then its

$$\boxed{U_{xx} + U_{yy} = 0}$$? then its a harmonic function

moving to the next stage

$\because V_y = U_x$ (CR eq^n rule)

do integration with respect to y

$$V(x,y) = (\sim\sim\sim) + g(x)$$

because x function got omitted in d/dy process

again,

doing

$$-V_x = \frac{\partial}{\partial x}(V(x,y)) = \underbrace{(\cdots)+g'(x)}_{\text{this is == to } U_y}$$

so $-V_x = U_y$

$\Rightarrow g(x) = (\cdots)$

do integration, this time not partial

$g(x) = (\cdots) + c$

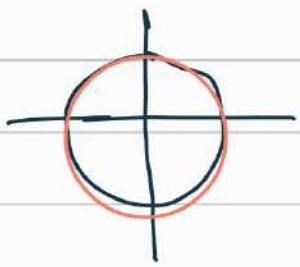
now plug this back

and I get the V

example
[slide pg 7]

$$V(x,y) = 2x + \frac{1}{x}\tan^{-1}\left(\frac{y}{x}\right) + g(x)$$

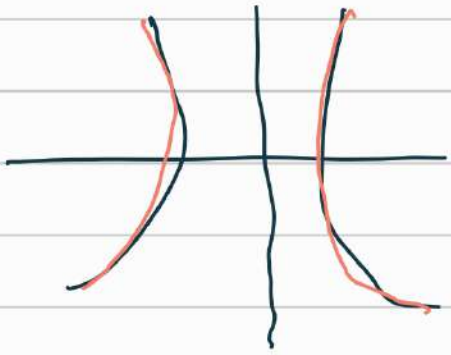
circular



$x^2+y^2=1$

$\sin(\theta)$ $\cos(\theta)$

hyprbolic



$x^2-y^2=1$

$\sinh(x)$ $\cosh(x)$

much simpler

#

$$\ln(Z) = \ln|Z| + i \cdot \underbrace{\arg(Z)}_{\text{Arg}Z + 2\pi n}$$

$$\ln(i^{1/2})$$

$$i = 1 \cdot e^{i\left(\frac{\pi}{2} + 2n\pi\right)}$$

$$\Rightarrow (i)^{1/2} = 1 \cdot e^{i\left(\pi/2 + 2n\pi\right)\frac{1}{2}}$$

$$\Rightarrow i^{1/2} = 1 \cdot e^{i\underbrace{\left(\frac{\pi}{4} + n\pi\right)}_{\arg Z}}$$

$$\Rightarrow \ln|i^{1/2}| = \underbrace{\cancel{\ln 1}}_{0} + i \cdot \left(\frac{\pi}{4} + n\pi\right)$$

[pg 10 of slide]

[pg 11 of slide]

[pg 11 of slide]

$\sin^{-1} z = w$, assume

$\sin w = z$

$$\Rightarrow \frac{e^{iw} - e^{-iw}}{2i} = z$$

$$\Rightarrow e^{iw} - e^{-iw} = 2zi$$

$$(e^{iw} + e^{-iw})^2$$

$$= (2iz)^2 + 4 \cdot e^{iw} \cdot e^{-iw}$$

$$= 4 - 4z^2$$

$$\Rightarrow e^{iw} + e^{-iw} = 2\sqrt{1-z^2}$$

$$2e^{iw} = 2zi + 2\sqrt{1-z^2}$$

$\sin(iz)$

$$= \frac{e^{i(iz)} - e^{-i(iz)}}{2i}$$

$$e^{-z} - e^{z}$$

$$= \frac{\quad}{2i}$$

$$= -i \, \frac{e^{-z} - e^{+z}}{2}$$

$$= i \, \frac{e^{z} - e^{-z}}{2}$$

$$= i \cdot \sinh(z)$$

$$\boxed{\text{lecture 10}}$$

Question will be hard this semester